

Complex Retrieval Queries using GROUP BY

Q. Explain the use of the GROUP BY clause in complex queries. Differentiate between WHERE and HAVING, and demonstrate the order of execution when combining GROUP BY, HAVING, and ORDER BY.

> **Definition to Remember**

> The
Function
similar

1. GROUP BY with Aggregate Functions

Instead of calculating an aggregate for the entire table, `GROUP BY` calculates it **per unique group**.

```
```sql
```

-- Find

### 2. HAVING Clause

`WHERE` cannot use aggregate functions. `HAVING` was introduced specifically to **filter groups** after `GROUP BY` executes.

```
```sql
```

-- Find

3. WHERE vs HAVING

| Feature | WHERE Clause | HAVING Clause |

| :--- | :---

4. SQL Query Execution Order

Understanding the strict order of execution is critical:

```
```
```

FROM

**Complex Example:**

sql

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> **Must-Write Points (for 10 marks)**

> 1. 'G

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> **Quick Recall**

> `GR  
result)